5G RAN

Flow Control Feature Parameter Description

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

The following describes the specific changes in different issues.

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-09-26).

- Technical changes

  None

- Editorial changes

  Corrected the control parameter of uplink user-plane flow control. For details, see **Uplink Flow Control**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FBFD-020101	Reliability	4 Control-Plane Flow Control 5 User-Plane Flow Control 6 Management-Plane Flow Control

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Control-plane flow control	Supported in both NSA and SA networking, with the following differences: ● Flow control over SgNB addition messages is supported only in NSA networking. ● AMF overload-triggered flow control, AC control, and flow control over random access messages, initial access messages, handover request messages, and paging messages are supported only in SA networking.	4 Control-Plane Flow Control
User-plane flow control	None	5 User-Plane Flow Control
Management-plane flow control	None	6 Management-Plane Flow Control

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Control-plane flow control	None	4 Control-Plane Flow Control

Function Name	Difference	Chapter/Section
User-plane flow control	None	**5 User-Plane Flow Control**
Management-plane flow control	None	**6 Management-Plane Flow Control**

3 Overview

3.1 Introduction

With flow control, a device controls input and output flows to prevent the device from being overloaded and maintain device stability. Flow control is performed on signaling, service, and operation and maintenance (O&M) data. The following two methods are used to ensure the effect of flow control:

- Input flows are restricted to prevent the device from being overloaded and ensure the processing capability of the device when its service traffic dramatically increases.
- Output flows are restricted to prevent the peer device from being overloaded.

When heavy traffic exists on the device, flow control can reduce the device reset risk and improve device reliability. Flow control also prevents the access success rate and handover success rate from deteriorating, ensuring user experience.

3.2 Application Scenarios

Control-plane, user-plane, and management-plane data flows exist on the network. These three types of data flows map to signaling, service, and O&M data of flow control objects, respectively, as shown in **Figure 3-1**.

Figure 3-1 gNodeB data flow

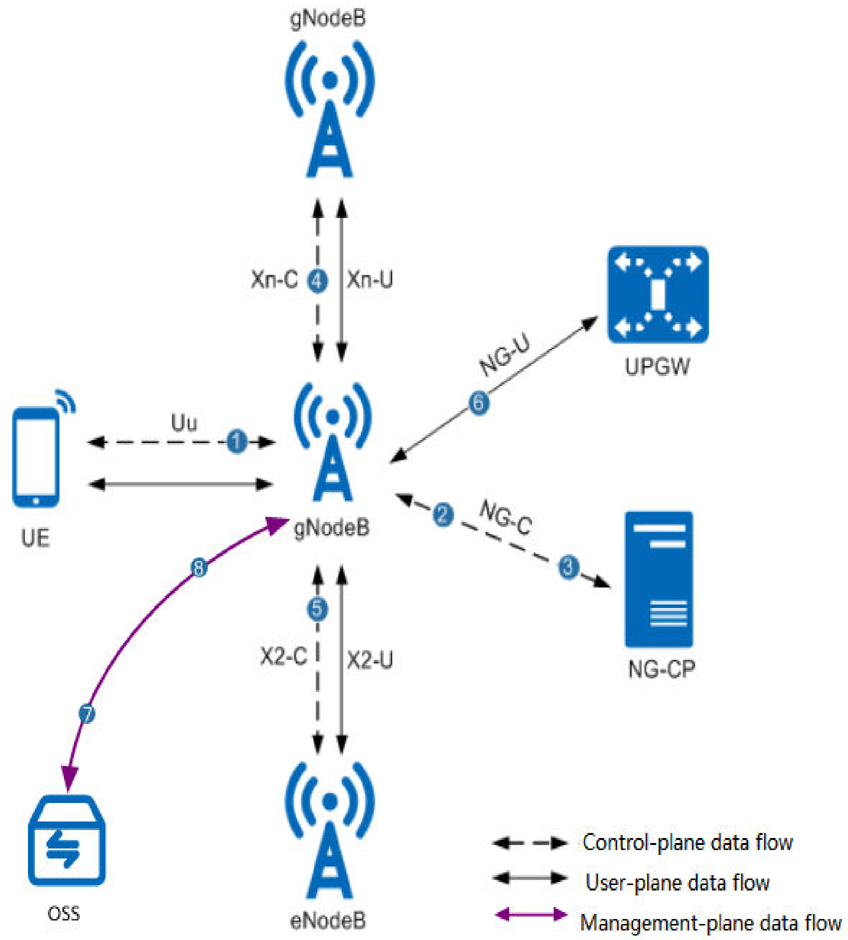


Table 3-1 Load points on NR networks

Data Flow Type	Data Flow	Load Point
Control-plane data flow	Uplink signaling data between a UE and a gNodeB	Load point 1: A gNodeB is overloaded if a UE sends a large amount of access signaling data over the Uu interface to the gNodeB.
	Downlink signaling data between a gNodeB and a next generation control plane (NG-CP)	Load point 2: A gNodeB is overloaded if an NG-CP sends a large amount of signaling data to the gNodeB.
	Uplink signaling data between a gNodeB and an NG-CP	Load point 3: An NG-CP is overloaded if a gNodeB sends a large amount of signaling data to the NG-CP.

Data Flow Type	Data Flow	Load Point
	Signaling data between gNodeBs	Load point 4: A gNodeB is overloaded if there is a large amount of signaling data between gNodeBs.
	Signaling data between a gNodeB and an eNodeB	Load point 5: A gNodeB is overloaded if there is a large amount of signaling data between a gNodeB and an eNodeB.
	Downlink signaling data between a gNodeB and a UE	There is no load point on the gNodeB side.
User-plane data flow	gNodeB uplink and downlink service data	Load point 6: The CPU of a gNodeB is overloaded if there is a large amount of uplink and downlink service data.
Manageme nt-plane data flow	Uplink OAM data flows from a gNodeB to the operations support system (OSS)	Load point 7: The OSS is overloaded if a gNodeB sends too much OAM data to the OSS.
	Downlink OAM data flows from the OSS to a gNodeB	Load point 8: A gNodeB is overloaded if the OSS sends too much OAM data to the gNodeB.

3.3 Basic Principles

Flow control is performed on control-plane and user-plane data flows within a gNodeB or between a gNodeB and an external NE.

Flow control methods are as follows:

- Restricting output flows of the gNodeB or reducing the data flows received from the peer NE through backpressure
- Actively reducing the flow output rate of the gNodeB, or that of the peer NE through backpressure
- Identifying service priorities and suppressing the access of low-priority data

NOTE

Backpressure is a traffic control method. When detecting that the transmit end transmits an excessively large volume of traffic, the receive end sends signals to instruct the transmit end to lower the transmission rate.

3.3.1 Control-Plane Data Flows and Protocols

As shown in **Figure 3-1**, control-plane data flows include:

- Uplink signaling data from a UE to a gNodeB
- Downlink signaling data from a gNodeB to a UE

- Uplink control-plane data from a gNodeB to an NG-CP
- Downlink signaling data from an NG-CP to a gNodeB
- Signaling data between gNodeBs
- Signaling data between a gNodeB and an eNodeB

Figure 3-2, **Figure 3-3**, and **Figure 3-4** show the protocol stacks related to the control plane.

Figure 3-2 Control-plane protocol stacks between a UE and an eNodeB and between an eNodeB and a gNodeB in NSA networking

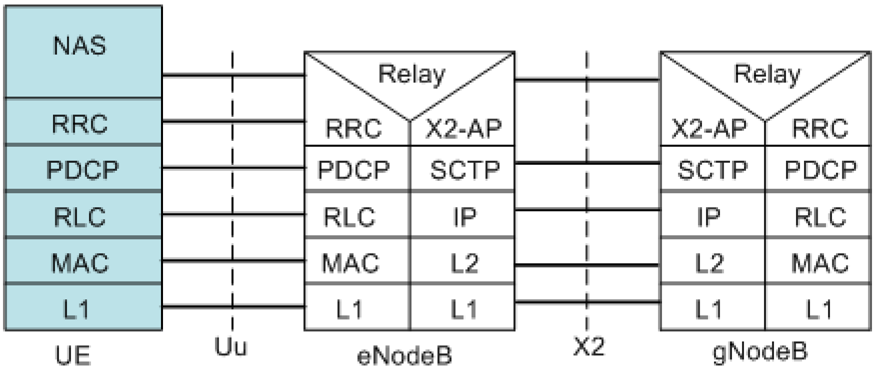


Figure 3-3 Control-plane protocol stacks between a UE and a gNodeB and between a gNodeB and an NGC in SA networking

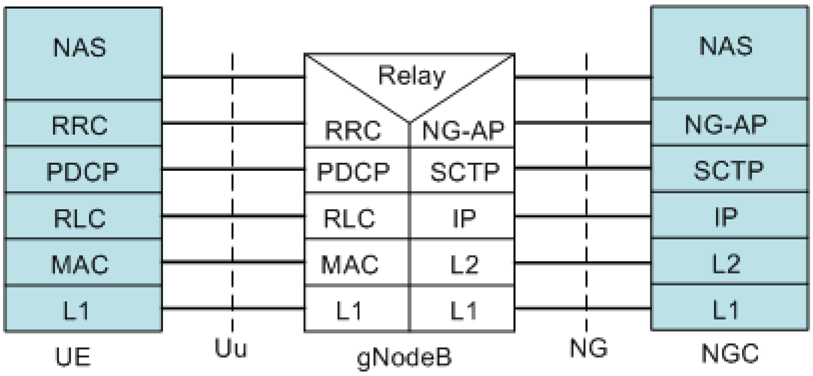
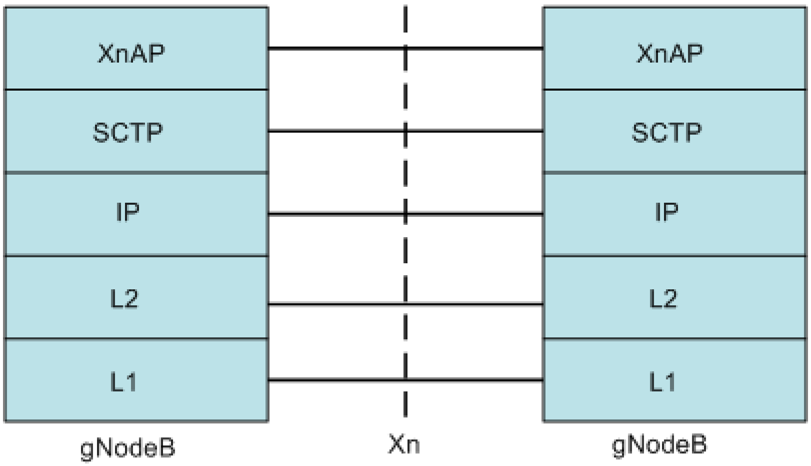


Figure 3-4 Control-plane protocol stacks between gNodeBs



3.3.2 User-Plane Data Flows and Protocols

As shown in **Figure 3-1**, user-plane data flows include:

- Uplink service data from a UE to a gNodeB
- Downlink service data from a gNodeB to a UE
- Service data between a gNodeB and an eNodeB
- Downlink service data from a user plane gateway (UPGW) to a gNodeB
- Uplink service data from a gNodeB to a UPGW
- Service data between gNodeBs

Figure 3-5, **Figure 3-6**, and **Figure 3-7** show the protocol stacks related to the user plane.

Figure 3-5 User-plane protocol stacks between a UE and a gNodeB, between a gNodeB and an eNodeB, and between an eNodeB and an S-GW in NSA networking

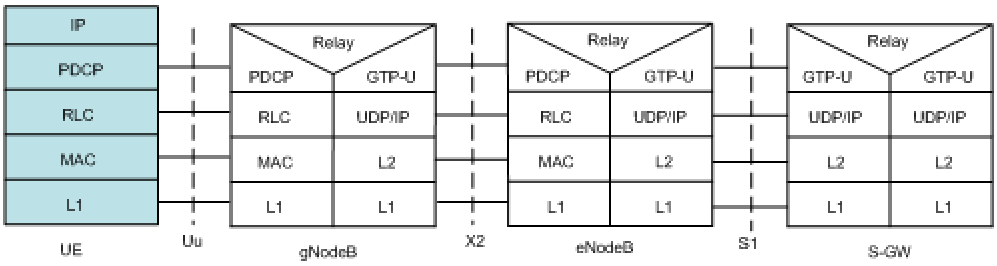


Figure 3-6 User-plane protocol stacks between a UE and a gNodeB and between a gNodeB and an NGC in SA networking

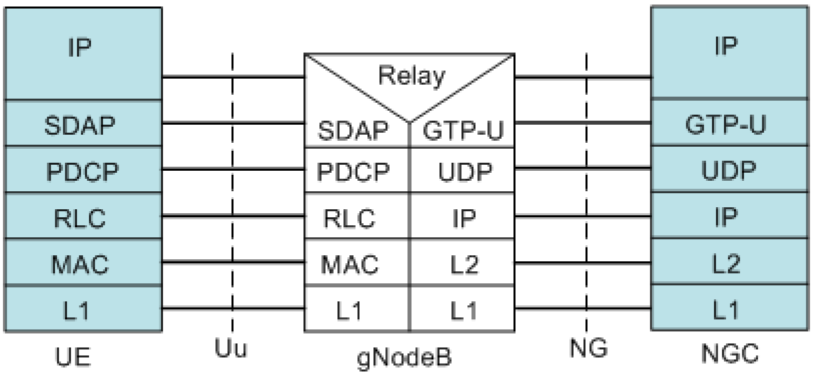
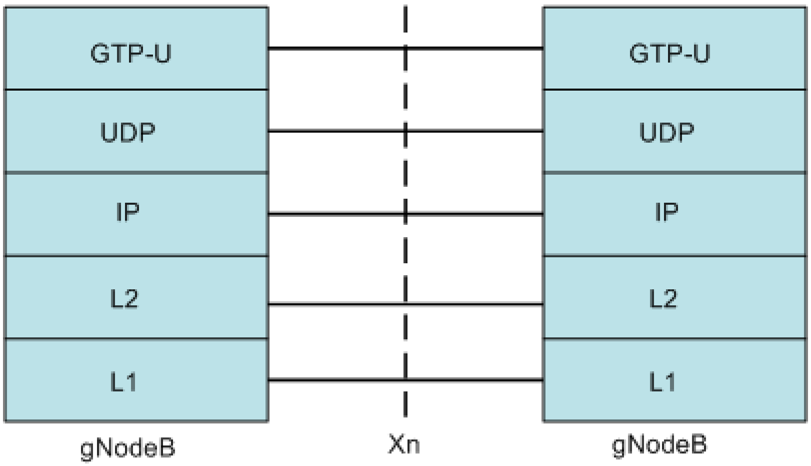


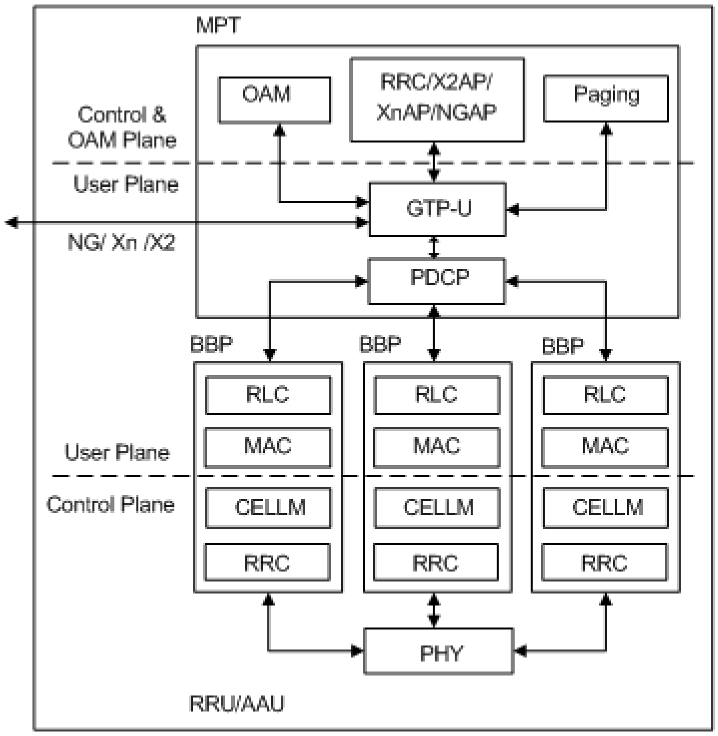
Figure 3-7 User-plane protocol stacks between gNodeBs



3.3.3 Internal Functional Architecture of a gNodeB

Figure 3-8 shows the internal functional architecture of a gNodeB.

Figure 3-8 Internal functional architecture of a gNodeB



Functions of the main processing and transmission unit (MPT) are as follows:

- Radio resource control (RRC), X2 Application Protocol (X2AP), Xn Application Protocol (XnAP), and NG Application Protocol (NGAP) functions and paging functions on the control plane
  - RRC functions over the Uu interface
  - X2AP functions over the X2 interface

  - XnAP functions over the Xn interface
  - NGAP functions over the NG interface
  - Paging functions
- Packet Data Convergence Protocol (PDCP) and GPRS Tunneling Protocol-User Plane (GTP-U) functions on the user plane:
  - PDCP functions over the Uu interface
  - GTP-U user plane functions and transmission algorithms over the X2, Xn, and NG interfaces
- OM functions in the OAM plane

Functions of the baseband processing unit (BBP) are as follows:

- RRC and Cell Manage (CELLM) functions on the control plane
  - RRC functions over the Uu interface
  - CELLM functions, including the Uu interface resource allocation function
- Radio Link Control (RLC) and Media Access Control (MAC) functions over the Uu interface on the user plane
- Functions in the OAM plane

4 Control-Plane Flow Control

4.1 Flow Control over SgNB Addition Messages

4.1.1 Objective

In NSA networking, the signaling data exchanged between a UE and a gNodeB is carried by an eNodeB. Therefore, control-plane flow control is only performed on the X2 interface between an eNodeB and a gNodeB to prevent signaling overload.

4.1.2 Principle

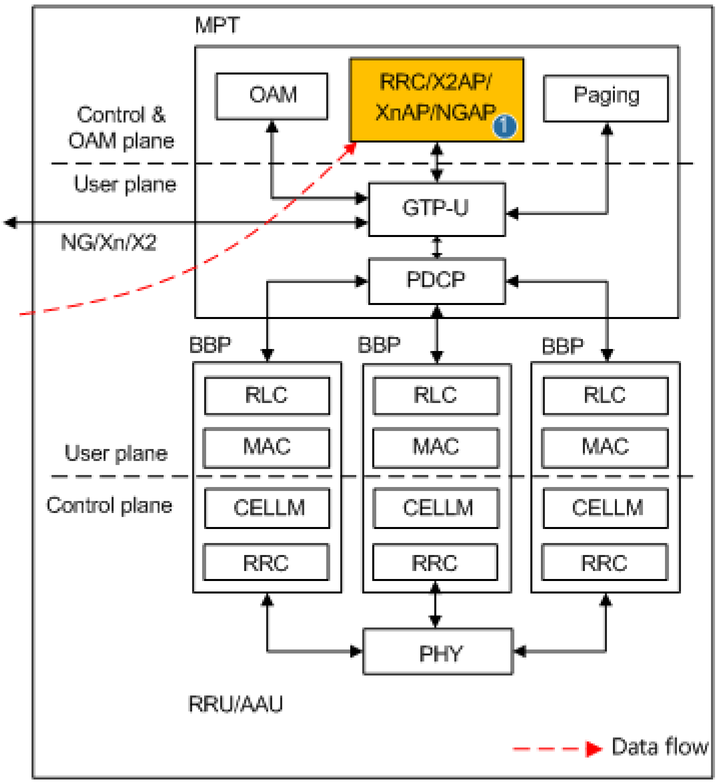
In an SgNB addition procedure initiated by the eNodeB over the X2 interface, the gNodeB informs the eNodeB of available radio resources through a configuration message over the X2 interface, and then the eNodeB informs the UE of the radio resources over the Uu interface. For details about the SgNB addition procedure, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*.

In this release, only the initial messages (SgNB ADDITION REQUEST messages) over the X2 interface between the eNodeB and gNodeB are subject to flow control.

4.1.2.1 Flow Control Points

Figure 4-1 shows the data flow for SgNB addition messages and the flow control point (marked 1) for initial messages (SgNB ADDITION REQUEST messages) in an SgNB addition procedure in the gNodeB. The gNodeB performs flow control on the initial messages to prevent signaling overload.

Figure 4-1 Data flow and flow control point for SgNB addition messages



4.1.2.2 Flow Control Actions

The MPT collects the number of SgNB addition messages received from the eNodeB within a period (one second), and determines whether this period has sufficient tokens for processing all the received SgNB addition messages (one token is consumed for processing one SgNB addition message). If this period does not have sufficient tokens, the gNodeB directly discards the SgNB addition messages that cannot be processed.

When the gNodeB is overloaded, the gNodeB dynamically adjusts the number of tokens used in the next one-second period based on the CPU usage of the MPT, message queue status on the MPT, or transmission link status. **Table 4-1** details actions for flow control over SgNB addition messages.

Table 4-1 Flow control over SgNB addition messages

MPT CPU Usage (*N*)	MPT Message Queue Status or Transmission Link Status	Flow Control Status	Flow Control Actions on the MPT
N ≥ 95%	-	Overload	The CPU load is relatively high. All SgNB addition messages will be discarded to prevent system breakdown. The number of tokens is decreased at a specified step within each period.
80% ≤ *N* < 95%	-	Flow control enabled	The number of tokens is decreased at a specified step within each period.
N < 80%	Congestion	Flow control enabled	The number of tokens is decreased at a specified step within each period.
N < 80%	Congestion clearance	Flow control released	The number of tokens is increased at a specified step within each period.
Note: The step is adjusted following the slow-increase and quick-decrease principle.			

4.1.3 Monitoring

View the value of **N.NsaDc.SgNB.Add.Disc.FlowCtrl** to check the number of times SCG addition messages are discarded due to the flow control over SgNB addition messages on the base station.

4.2 AMF Overload-triggered Flow Control

4.2.1 Objective

The objective of AMF overload-triggered flow control is to relieve the impact of AMF overload caused by a large number of UEs' access to the network. This scenario corresponds to load point 3 in **Figure 3-1**.

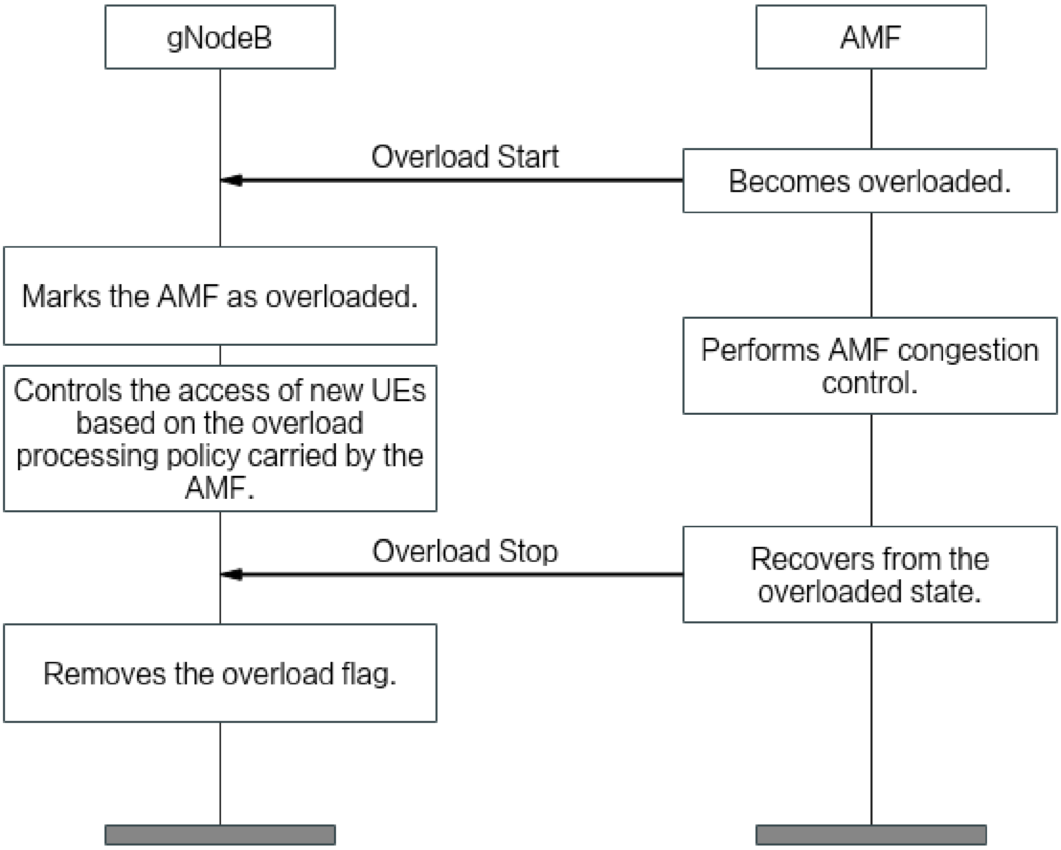
4.2.2 Principle

When an AMF is overloaded, it performs congestion control and sends an Overload Start message to notify the gNodeB. Upon receipt of this message, the gNodeB marks the AMF as overloaded. In the Overload Start message, the AMF instructs the gNodeB to allow or reject calls of UEs of a specific type and indicates the proportion of UEs allowed or rejected. The gNodeB controls whether a UE can access the overloaded AMF based on the indication.

When the AMF is no longer overloaded, it sends an Overload Stop message to the gNodeB. After receiving this message, the gNodeB updates the overload flag of the AMF and the corresponding AMF set information, and resumes normal processing.

Figure 4-2 shows the principle of flow control triggered by AMF overload. For details about AMF overload handling, see *NG-flex*.

Figure 4-2 Principle of flow control triggered by AMF overload



4.2.3 Monitoring

None

4.3 Flow Control over Random Access Messages

4.3.1 Objective

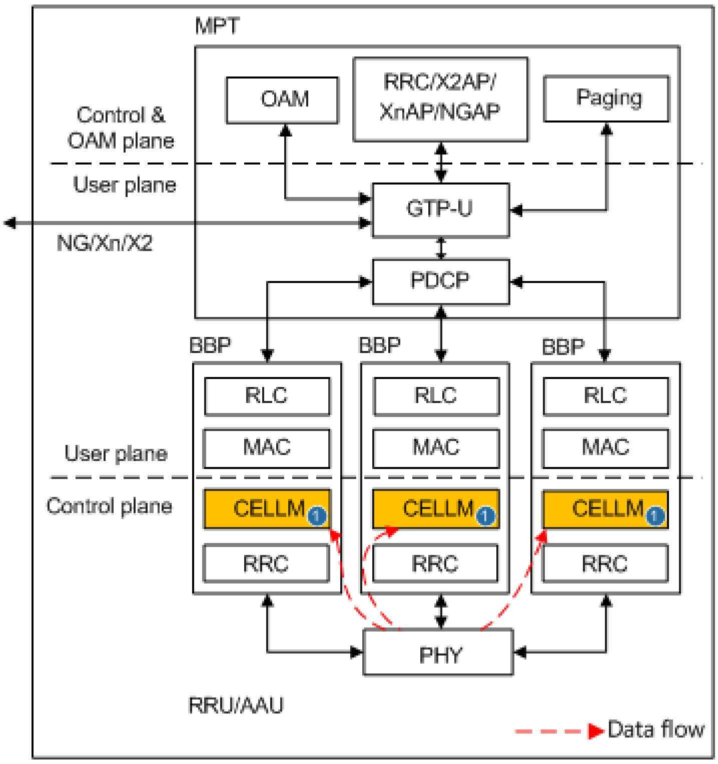
A large number of random access messages lead to high load and even cause the system to reset. Flow control over random access messages aims to protect a gNodeB from being overloaded when a large number of UEs initiate random access.

4.3.2 Principle

4.3.2.1 Flow Control Points

Figure 4-3 shows the data flow for random access messages and the flow control point (located in the CELLM processing module of the BBP, as marked 1) for contention-based random access messages (Msg1) in the gNodeB. The gNodeB does not perform flow control on non-contention-based random access messages.

Figure 4-3 Data flow and flow control point for random access messages



4.3.2.2 Flow Control Actions

Flow control over contention-based random access messages is implemented on the BBP. After receiving Msg1 from a UE, the BBP uses a group of instance resources from those allocated by the MPT to send Msg2 (random access response message) to the UE. The MPT obtains the CPU usage every second. If the MPT CPU usage exceeds 85%, the MPT message queue is congested, or the transmission link is congested, backpressure is performed on the BBP to limit the number of Msg1 messages. In other cases, flow control over random access messages is released and the MPT notifies the BBP to restore normal services. The BBP implements flow control over random access messages every 100 ms, and adjusts the number of tokens within each flow control period. One token is consumed each time the gNodeB processes one Msg1. If the number of Msg1 messages to be processed in the current period exceeds the total number of tokens, new Msg1 messages will be discarded. **Table 4-2** details flow control actions. The gNodeB determines whether to trigger flow control based on the backpressure status on the MPT.

Table 4-2 Actions for flow control over random access messages

MPT CPU Usage (*N*)	MPT Message Queue Status or Transmission Link Status	Flow Control Status	Flow Control Actions on the BBP
N ≥ 95%	-	Overload	The CPU load is relatively high. All random access messages will be directly discarded to prevent system breakdown. The number of tokens is decreased at a specified step within each period.
85% ≤ *N* < 95%	-	Flow control enabled	The number of tokens is decreased at a specified step within each period.
N < 85%	Congestion	Flow control enabled	The number of tokens is decreased at a specified step within each period.
N < 85%	Congestion clearance	Flow control released	The number of tokens is increased at a specified step within each period.
Note: The step is adjusted following the slow-increase and quick-decrease principle.			

4.3.3 Monitoring

None

4.4 Flow Control over Initial Access Request Messages

4.4.1 Objective

A large number of initial access request messages lead to high load and even cause the system to reset. Flow control over initial access request messages (Msg3) aims to relieve the gNodeB overload caused by the access of a large number of UEs.

4.4.2 Principle

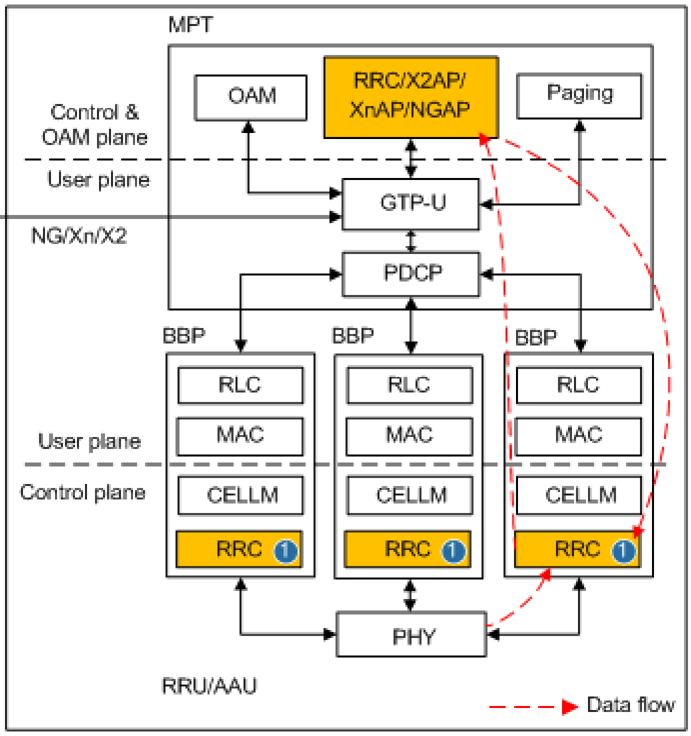
An RRC procedure starts with an initial access request message. Initial access request messages include RRC Setup Request and RRC Reestablishment Request messages. After an initial access request message is successfully processed, a series of subsequent operations are required, causing excessive signaling overheads on the system. Therefore, flow control is necessary from the beginning of a signaling procedure to reduce the system load.

To guarantee user experience for high-priority services, control-plane flow control takes different actions to process access requests based on the priorities of access with different causes. The priorities for control-plane access requests are arranged in descending order of the causes: Emergency, highPriorityAccess concerns, mt-Access, Mo-Sig, Mo-Data, RRCReestablishment, and other RRC causes.

4.4.2.1 Flow Control Points

Figure 4-4 shows the data flow and flow control point (located in the RRC processing module of the BBP, as marked 1) for initial access request messages in the gNodeB.

Figure 4-4 Data flow and flow control point for initial access request messages



4.4.2.2 Flow Control Actions

The BBP collects the number of initial access messages received from UEs within a period (one second), and determines whether this period has sufficient tokens for processing all the received RRC request messages (one token is consumed for processing one RRC request message).

When the gNodeB is overloaded, the gNodeB discards some initial access messages based on the CPU usage of the MPT or BBP, the message queue status of the MPT or BBP, or the transmission link status. **Table 4-3** lists detailed flow control measures.

Table 4-3 Actions for flow control over initial access request messages

MPT/BBP CPU Usage (*N*)	MPT/BBP Message Queue Status or Transmission Link Status	Flow Control Status	Flow Control Actions on the BBP
N ≥ 95%	-	Overload	The CPU load is relatively high. All initial access messages will be discarded to prevent system breakdown. The number of tokens is decreased at a specified step within each period.
80% ≤ *N* < 95%	-	Flow control enabled	Flow control is performed on all types of messages according to the priorities for access causes of control-plane access requests, and the number of tokens is decreased at a certain step within each period. Messages of rejected UEs will be directly discarded.
N < 80%	Congestion	Flow control enabled	Flow control is performed on all types of messages according to the priorities for access causes of control-plane access requests, and the number of tokens is decreased at a certain step within each period. Messages of rejected UEs will be directly discarded.
N < 80%	Congestion clearance	Flow control released	Flow control is gradually released for all types of messages according to the priorities for access causes of control-plane access requests, and the number of tokens is increased at a certain step within each period.
Note: The step is adjusted following the slow-increase and quick-decrease principle.			

4.4.3 Monitoring

- View the value of **N.RRC.SetupReq.Msg.Disc.FlowCtrl** to check the number of discarded RRC Setup Request messages due to flow control over initial access request messages in SA networking.
- View the value of **N.RRC.ReEst.Msg.Disc.FlowCtrl** to check the number of discarded RRC Reestablishment Request messages due to flow control over initial access request messages in SA networking.

4.5 Flow Control over Handover Request Messages

4.5.1 Objective

In SA networking, a handover procedure starts with a handover request message (HANDOVER REQUEST). After a handover request message is successfully processed, a series of subsequent operations are required, causing excessive overheads on the system. Therefore, flow control over handover request messages needs to be implemented from the beginning of a signaling procedure to reduce system load.

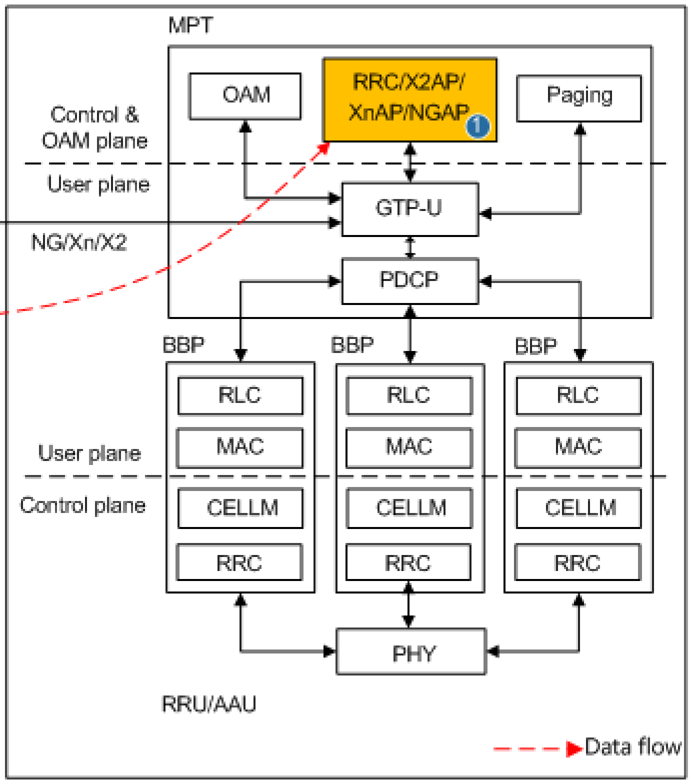
4.5.2 Principle

This release allows for flow control over handover request messages (initial messages in an Xn or NG handover procedure). If the CPU load on the gNodeB exceeds a flow control threshold within a period, the gNodeB performs flow control over handover request messages by lowering the capability of processing handover request messages. When the number of handover request messages is beyond the message processing capability within this period, the gNodeB will directly discard the handover request messages that cannot be processed.

4.5.2.1 Flow Control Points

Figure 4-5 shows the data flow and flow control point (located in the RRC/X2AP/XnAP/NGAP processing module of the MPT board, as indicated by "1" in the figure) for handover request messages in the gNodeB. The same flow control priority applies to handover request messages over the NG and Xn interfaces.

Figure 4-5 Data flow and flow control point for handover request messages



4.5.2.2 Flow Control Actions

The MPT collects the number of handover request messages received from the gNodeB or core network within a period (one second), and determines whether this period has sufficient tokens for processing all of the received handover request messages (one token is consumed for processing one handover request message). If this period does not have sufficient tokens, the gNodeB directly discards the handover request messages that cannot be processed.

When the gNodeB is overloaded, the gNodeB dynamically adjusts the number of tokens used in the next one-second period based on the CPU usage of the MPT, message queue status on the MPT, or transmission link status. **Table 4-4** details actions for flow control over handover request messages.

Table 4-4 Actions for flow control over handover request messages

MPT CPU Usage (*N*)	MPT Message Queue Status or Transmission Link Status	Flow Control Status	Flow Control Actions on the MPT
N ≥ 95%	-	Overload	The CPU load is relatively high. All handover request messages will be discarded to prevent system breakdown. The number of tokens is decreased at a specified step within each period.
90% ≤ *N* < 95%	-	Flow control enabled	The number of tokens is decreased at a specified step within each period.
N < 90%	Congestion	Flow control enabled	The number of tokens is decreased at a specified step within each period.
N < 90%	Congestion clearance	Flow control released	The number of tokens is increased at a specified step within each period.
Note: The step is adjusted following the slow-increase and quick-decrease principle.			

4.5.3 Monitoring

- View the value of **N.HO.IntraRAT.Msg.Disc.FlowCtrl** to check the number of times intra-RAT incoming handover messages are discarded due to the flow control over handover request messages in SA networking.
- View the value of **N.HO.InterRAT.Msg.Disc.FlowCtrl** to check the number of times inter-RAT incoming handover messages are discarded due to the flow control over handover request messages in SA networking.

4.6 Flow Control over Paging Messages

4.6.1 Objective

In SA networking, a large number of paging messages cause heavy system load and even a system reset. The aim of flow control over paging messages is to protect a gNodeB from being overloaded due to numerous paging messages.

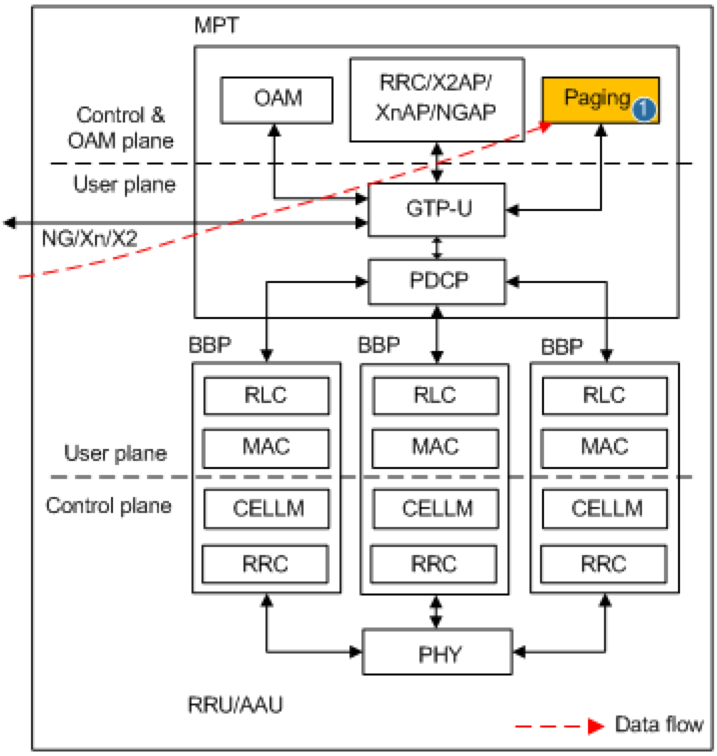
4.6.2 Principle

A paging procedure starts with a paging message. After a paging message is successfully processed, numerous UEs access the network, causing excessive overheads on the system. Therefore, flow control needs to be implemented from the beginning of a paging procedure to reduce system load.

4.6.2.1 Flow Control Points

Figure 4-6 shows the data flow and flow control point (located in the paging processing module of the MPT board, as indicated by "1" in the figure) for paging messages in the gNodeB. The same flow control priority applies to paging messages over the NG and Xn interfaces.

Figure 4-6 Data flow and flow control point for paging messages



4.6.2.2 Flow Control Actions

If the MPT CPU usage of the gNodeB exceeds 75%, the MPT message queue status is congested, or the transmission link status is congested within a period, the gNodeB performs flow control over paging messages by lowering the capability of processing paging messages. When the number of paging messages is beyond the message processing capability within this period, the gNodeB will directly discard the paging messages that cannot be processed.

The MPT collects the number of paging messages received from the gNodeB or core network within a one-second period, and determines whether tokens in this period are sufficient for processing all of the received paging messages (one token is consumed for processing one paging message). If tokens within this period are insufficient, the gNodeB directly discards the paging messages that cannot be processed.

When the **PAGING_CAPACITY_ENH_SW** option of the **gNodeBAlgo**.***PagingAlgoSwitch*** parameter is selected, the flow control specifications of the UMPTe/UMPTga in NR single-mode scenarios can be improved. However, the flow control specifications of the UMPTg meet the requirements and do not require improvement. For a board other than the UMPTe/UMPTga/UMPTg, its flow control specifications cannot be improved even though this option is selected.

If the gNodeB is overloaded, it dynamically adjusts the number of tokens used in the subsequent one-second period based on the CPU usage or message queue status on the MPT. **Table 4-5** details actions for flow control over paging messages.

Table 4-5 Actions for flow control over paging messages

MPT CPU Usage (*N*)	MPT Message Queue Status or Transmission Link Status	Flow Control Status	Flow Control Actions on the MPT
N ≥ 95%	-	Overload	The CPU load is relatively high. All paging messages will be discarded to prevent system breakdown. The number of tokens is decreased at a specified step within each period.
75% ≤ *N* < 95%	-	Flow control enabled	The number of tokens is decreased at a specified step within each period.
N < 75%	Congestion	Flow control enabled	The number of tokens is decreased at a specified step within each period.
N < 75%	Congestion clearance	Flow control released	The number of tokens is increased at a specified step within each period.
Note: The step is adjusted following the slow-increase and quick-decrease principle.			

4.6.3 Monitoring

- View the value of **N.Paging.Xn.Msg.Disc.FlowCtrl** to check the number of times paging messages over the Xn interface are discarded due to the flow control over paging messages in SA networking.
- View the value of **N.Paging.Ng.Msg.Disc.FlowCtrl** to check the number of times paging messages over the NG interface are discarded due to the flow control over paging messages in SA networking.

4.7 AC Control

4.7.1 Objective

Network congestion may occur when a large number of UEs access the network simultaneously. The purpose of AC control is to perform unified access control on UEs in idle or inactive state in a cell to reduce network congestion. This scenario corresponds to load point 1 in **Figure 3-1**.

4.7.2 Principle

AC control is implemented jointly by the gNodeB and UE. The procedure is as follows:

1. The gNodeB broadcasts access control information in SIB1 using any of the following methods:
   - Static AC control: Access control parameters are configured using MML commands and then broadcast to UEs through SIB1.
   - Dynamic AC control: Access control parameters are configured using MML commands, and the broadcast of related access control information to UEs through SIB1 is dynamically controlled based on the network load.
2. Before initiating an RRC connection setup or RRC connection resume procedure, a UE determines whether it can access the cell based on the access control information broadcast in SIB1.

   According to section 4.5 "Unified access control" in 3GPP TS 24.501 V16.5.0, when a UE needs to initiate an access attempt, it determines the corresponding access ID and access category, checks SIB1 for the broadcast access control information (including the access ID and access category), and determines whether access is allowed accordingly. If access is allowed, the UE initiates an access procedure. If not allowed, the UE does not initiate an access procedure. For details about the procedure for a UE to determine whether its access attempt is allowed, see section 5.3.14 "Unified Access Control" in 3GPP TS 38.331 V15.6.0.

For details about AC control, see *Access Class Control*.

4.7.3 Monitoring

1. To check whether AC control has taken effect, perform the following steps:
   a. Log in to the MAE-Access, and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **Trace Type** > **NR** > **Application Layer** > **Uu Interface Trace**.
   b. Check whether the *uac-BarringInfo* IE is included in SIB1. If included, the AC control function has taken effect.
2. Run the **DSP NRDUCELLUACBAR** command to query the effective state of AC control.
   - If the value of **Flow Control-based UAC State** or **RRC_CONNECTED UE Number-based UAC State** in **Access Control State** is **On** in the command output, the dynamic AC control function has taken effect.

   - If the value of **STATIC_UAC** in **Access Control State** is **On** in the command output, the static AC control function has taken effect.
3. Observe the following performance counters to check whether dynamic AC control has taken effect:
   - If the value of the **N.DynACBar.Dur** counter is not 0, the dynamic AC control function has taken effect in the cell.
   - If the value of the **N.DynACBar.Dur.PLMN** counter is not 0, the dynamic AC control function has taken effect for the operator of the specified PLMN in the cell.

5 User-Plane Flow Control

5.1 Objective

The RLC and MAC protocol processing of a gNodeB and air interface transmission must meet specific delay requirements. If the CPU is overloaded, a gNodeB will be unable to meet delay requirements for air interface transmission, resulting in packet loss. Therefore, load control needs to be performed based on the CPU usage to ensure real-time RLC and MAC protocol processing.

5.2 Principle

The RLC and MAC protocol processing units of a gNodeB periodically check the CPU load and calculate the load from each cell. If the CPU load exceeds the flow control threshold, flow control is performed on the cell with the highest load by reducing the number of scheduled users and traffic. After the number of users scheduled for the cell is reduced, the RLC buffer may increase. In this case, congestion backpressure actions are taken by instructing the transmission module to reduce the number of transmitted packets. This helps avoid a buffer overflow and alleviates CPU load.

In this release, flow control in the uplink and downlink is supported.

5.2.1 Flow Control Actions

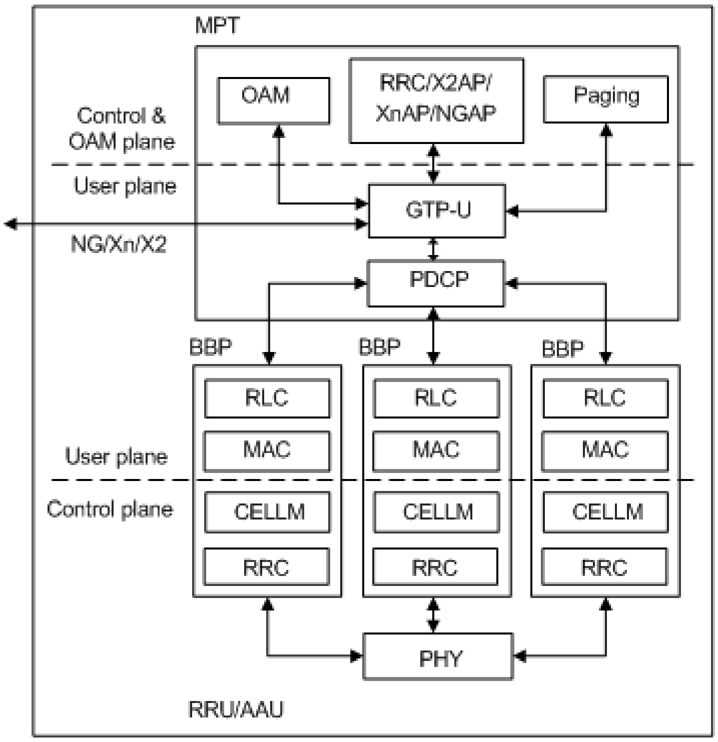
Downlink Flow Control

The downlink traffic flows among the protocol processing units of the gNodeB in the following direction: GTP-U > PDCP > RLC > MAC, as shown in **Figure 5-1**. The downlink flow control on the user plane is performed on the RLC and MAC protocol processing units.

- The RLC protocol processing unit receives and buffers packets from the PDCP processing unit. If the CPU load of the RLC and MAC protocol processing units exceeds 85%, the GTP-U module is instructed to lower the transmit rate for downlink packets until the CPU load is below 80%. Afterwards, the GTP-U module is instructed to gradually restore the transmit rate for downlink packets.

- The RLC and MAC protocol processing units process the downlink packets as instructed by the downlink scheduling module.
  - If the CPU load of the RLC and MAC protocol processing units exceeds 80%, the downlink scheduling module is instructed to reduce the number of users scheduled in the corresponding cell in the downlink. This helps reduce the CPU load and ensure real-time RLC and MAC protocol processing.
  - If the CPU load of the RLC and MAC protocol processing units is less than 80%, the congestion has been relieved. In this case, the downlink scheduling module is notified to increase the number of users scheduled in the downlink in the corresponding cell.

Figure 5-1 Functional architecture of a gNodeB



Uplink Flow Control

The uplink traffic flows among the protocol processing units of the gNodeB in the following direction: MAC > RLC > PDCP > GTP-U, as shown in **Figure 5-1**. The uplink flow control on the user plane is performed on the MAC, RLC, and PDCP protocol processing units.

- After being overloaded, the GTP-U processing unit notifies the PDCP processing unit of the overload through backpressure. Then, the PDCP processing unit decreases the uplink packet transmit rate.
- The RLC and MAC protocol processing units process the uplink packets as instructed by the uplink scheduling module.
  - If the CPU load of the RLC and MAC protocol processing units exceeds 80%, the uplink scheduling module is instructed to reduce the number of users scheduled in the corresponding cell in the uplink. This helps reduce the CPU load and ensure real-time RLC and MAC protocol processing.

– If the CPU load of the RLC and MAC protocol processing units is less than 80%, the congestion has been relieved. In this case, the uplink scheduling module is notified to increase the number of users scheduled in the corresponding cell in the uplink.

Uplink user-plane flow control is controlled by the **UL_UU_FLOW_CTRL_SW** option (non-dual connectivity (DC) scenarios) or **UL_UU_FLOW_CTRL_ENHANCED_SW** option (DC scenarios) of the **gNodeBParam**.***UlFlowCtrlAlgoSwitch*** parameter. When the specific option is selected, the MAC processing unit of the BBP performs uplink user-plane flow control to limit the uplink rates of UEs in case of uplink transmission congestions, thereby avoiding gNodeB internal congestions in the uplink.

- When the gNodeB is overloaded, the BBP reduces available resources at a step of 10%. The MAC protocol processing unit schedules a small amount of data in the uplink, accordingly.
- When the gNodeB is no longer overloaded, the BBP increases available resources at a step of 10%. The MAC protocol processing unit schedules a large amount of data in the uplink, accordingly.

NOTE

The **UL_UU_FLOW_CTRL_ENHANCED_SW** option can be selected only when the **UL_UU_FLOW_CTRL_SW** option is selected.

5.3 Monitoring

The trigger of user-plane flow control cannot be observed based on traffic-related counters because traffic change is related to various factors, such as normal traffic volume fluctuation and channel quality over the air interface. However, the counters indicating CPU usage of the BBP can be used to evaluate whether user-plane flow control has been triggered. The average user-plane CPU usage of a board in a gNodeB is indicated by the **VS.NRBoard.UPlane.CPULoad.Avg** counter. If the average user-plane CPU usage increases to about 80% and then remains this level, flow control has taken effect.

6 Management-Plane Flow Control

The management plane (OAM plane) handles device configuration and maintenance, involving tasks such as software upgrades and upload and download of a large number of files. This increases the CPU usage of the gNodeB and affects transmission of signaling messages in the control plane and data in the user plane during peak hours. The priorities of most OAM tasks are lower than those of control-plane and user-plane data transmission. OAM-plane flow control releases CPU resources, which will be used for control-plane and user-plane data transmission.

This chapter describes the objectives, principles, and monitoring methods of OAM-plane flow control.

6.1 OAM Flow Control

6.1.1 Objective

OAM flow control aims to prevent OAM tasks from occupying too many resources and negatively affecting control-plane and user-plane data transmission. OAM flow control corresponds to load points 7 and 8 in **Figure 3-1**.

6.1.2 Principle

When the CPU usage of a gNodeB is high, the gNodeB performs flow control over tasks of log recording, software management, and performance monitoring based on service priorities to ensure that key performance indicators (KPIs) of services do not deteriorate.

The priorities of control-plane, user-plane, and OAM-plane services are described as follows:

1. High-priority OAM tasks, including tasks related to MML commands, alarms, performance counters, and high-priority logs
2. Control-plane and user-plane data transmission
3. Medium-priority OAM tasks, including tasks related to medium-priority logs
4. Low-priority OAM tasks, such as tasks related to low-priority logs, radio access network (RAN) trace and monitoring, and software download

NOTE

- For details about the RAN trace and monitoring tasks, such as **Cell Performance Monitoring**, **User Performance Monitoring**, **Device Monitoring**, **Transport Performance Monitoring**, and **Information Collection**, see "Reference for Radio Access Network Trace or Monitoring Interfaces" in *MAE-Access MBB Network Management System Product Documentation*.
- High-, medium-, and low-priority logs are defined as follows:
  - High-priority logs include security, running, user operation, and traffic statistics logs.
  - Medium-priority logs include CHR, SON, SIG, TRC, and ECR logs.
  - Low-priority logs refer to debug logs.

If the CPU of the gNodeB is not overloaded but the data volume reported by the RAN trace task per second exceeds the processing threshold, the excess data will be discarded.

When the CPU of the gNodeB is overloaded, the OAM flow control is performed as follows:

- The gNodeB does not perform flow control over high-priority OAM tasks.
- The gNodeB rejects new trace and monitoring tasks and deletes existing trace and monitoring tasks.

  If the CPU usage of the MPT or BBP of the gNodeB is higher than 70% for 5 consecutive seconds, the gNodeB rejects new RAN trace and monitoring tasks and deletes existing RAN trace and monitoring tasks.
- The gNodeB stops recording logs based on log priorities.
  - When the CPU usage of the MPT or BBP of the gNodeB is greater than 70% for 5 consecutive seconds, the gNodeB stops recording low-priority logs.
  - When the CPU usage of the MPT or BBP of the gNodeB is greater than 80% for 5 consecutive seconds, the gNodeB stops recording medium-priority logs.

When the CPU usage of the gNodeB decreases, the OAM flow control recovery mechanism is as follows:

- The gNodeB allows new trace and monitoring tasks.

  When the CPU usage of the MPT or BBP of the gNodeB is less than 70% for 5 consecutive seconds, the gNodeB allows new RAN trace and monitoring tasks.
- The gNodeB resumes log recording based on log priorities.
  - When the CPU usage of the MPT or BBP of the gNodeB is less than 80% for 5 consecutive seconds, the gNodeB resumes recording medium-priority logs.
  - When the CPU usage of the MPT or BBP of the gNodeB is less than 70% for 5 consecutive seconds, the gNodeB resumes recording low-priority logs.

The gNodeB triggers MR flow control as follows:

- MR flow control triggered based on the CPU load: When numerous UEs report MRs simultaneously, a large number of gNodeB system resources are occupied, increasing the CPU load. In this case, the gNodeB controls the number of MRs reported per second to reduce gNodeB resource consumption and ensure user experience.

- MR flow control triggered based on the number of RRC_CONNECTED UEs in a cell: When the number of RRC_CONNECTED UEs in a cell is greater than the value of **NRCellOamParam**.***SigChrCtrlStateUeNumThld***, the cell enters the MR flow control state. In this case, the maximum number of UEs that can report MRs cannot exceed the value of the **NRCellOamParam**.***SigChrCtrlStateSelMaxUeNum*** parameter.

Trace reporting flow control triggered based on the CPU load: When numerous UEs report trace signaling simultaneously, a large number of gNodeB system resources are occupied, increasing the CPU load. When the CPU load of the main control board is higher than 60%, the gNodeB does not record trace signaling to reduce the resource consumption and ensure user experience.

NOTE

If the **NRCellOamParam**.***SigChrCtrlStateUeNumThld*** parameter is set to **65535**, SIG CHR flow control based on the number of RRC_CONNECTED UEs does not take effect.

If the **NRCellOamParam**.***SigChrCtrlStateUeNumThld*** parameter is set to **0**, periodic SIG CHR reporting is not performed. This parameter is valid only when it is set to a value other than **65535**.

6.1.3 Monitoring

OAM flow control interrupts performance monitoring tasks. A message is displayed on the MAE-Access indicating that the task has stopped because the gNodeB is performing flow control.

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"

3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"

3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3"

EPC-based NSA Basics

EPC-based NSA Performance Enhancement

EPC-based NSA Experience Improvement

NG-flex

Access Class Control

MAE-Access MBB Network Management System Product Documentation

